

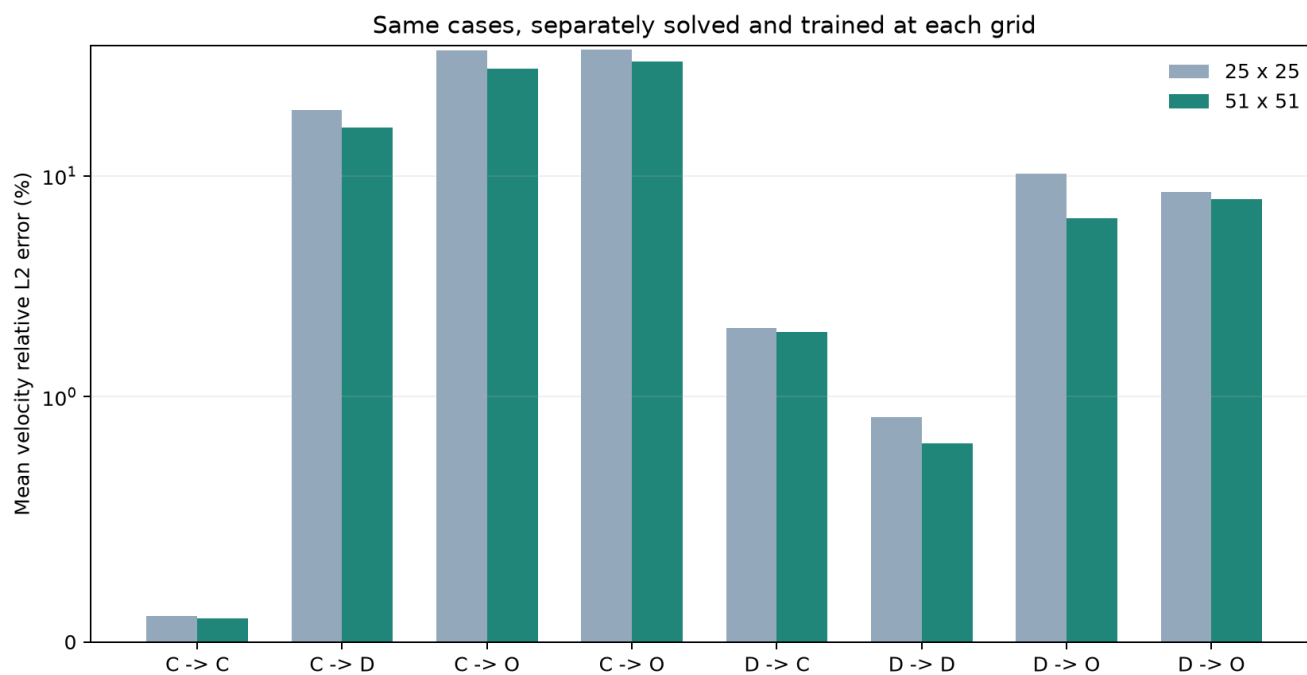
51 x 51 numerical refinement

Week 4.2 | 1/3

Matched Stokes fields and Navier-Stokes labels were solved again on 51 x 51 nodes.

The retained case manifest is unchanged; tests are regression cases, not fresh blind data.

The selected POD-MLP architecture was retrained from scratch at 51 x 51.

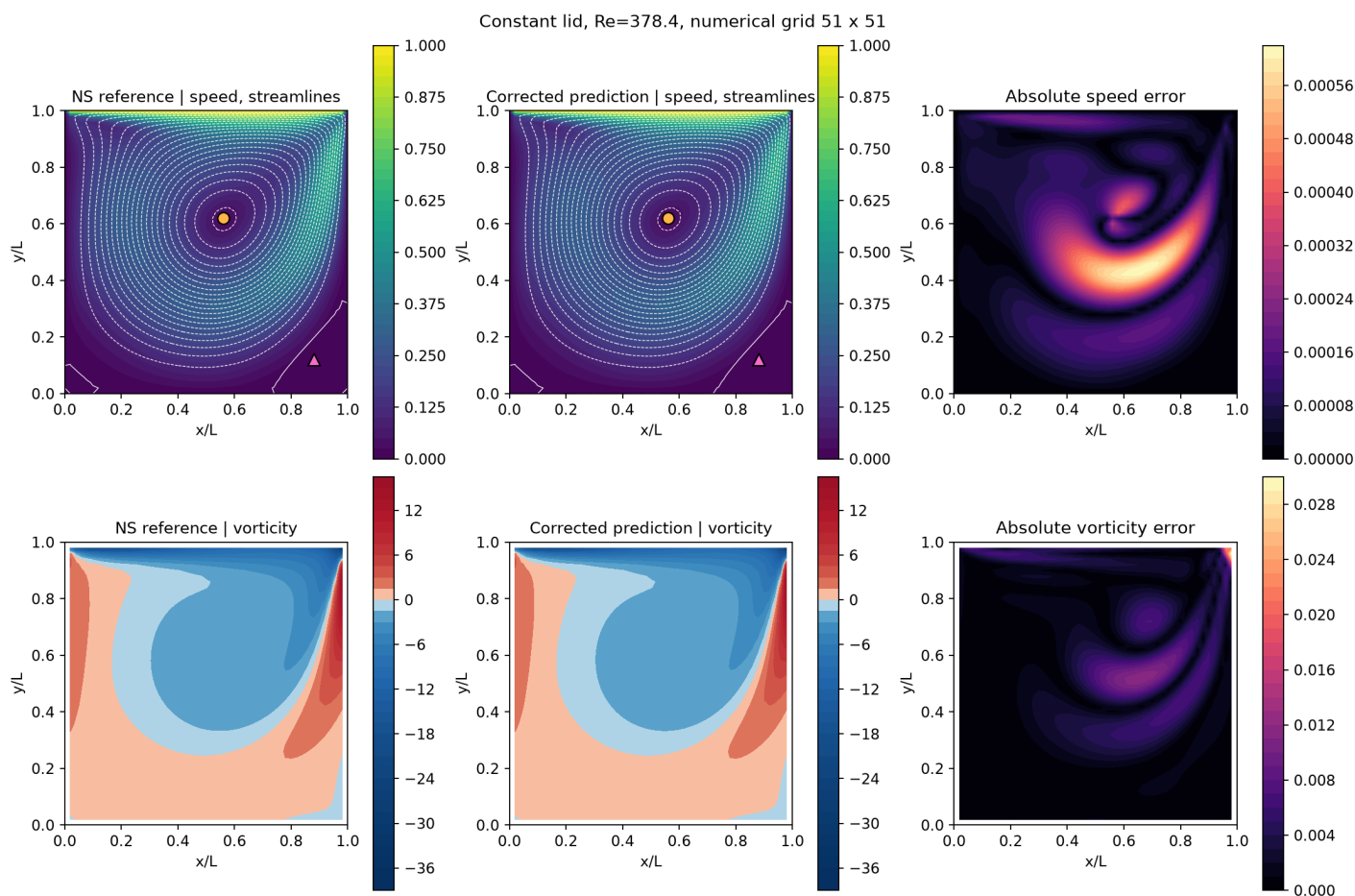


Train -> test	25 x 25	51 x 51
constant -> constant	0.107%	0.097%
constant -> diverse	19.920%	16.555%
constant -> ood_re	36.982%	30.700%
constant -> ood_shape	37.192%	33.068%
diverse -> constant	2.044%	1.960%
diverse -> diverse	0.917%	0.810%
diverse -> ood_re	10.204%	6.425%
diverse -> ood_shape	8.428%	7.878%

Errors compare models with separate numerical labels on each grid.

Constant lid: speed, streamlines, vorticity

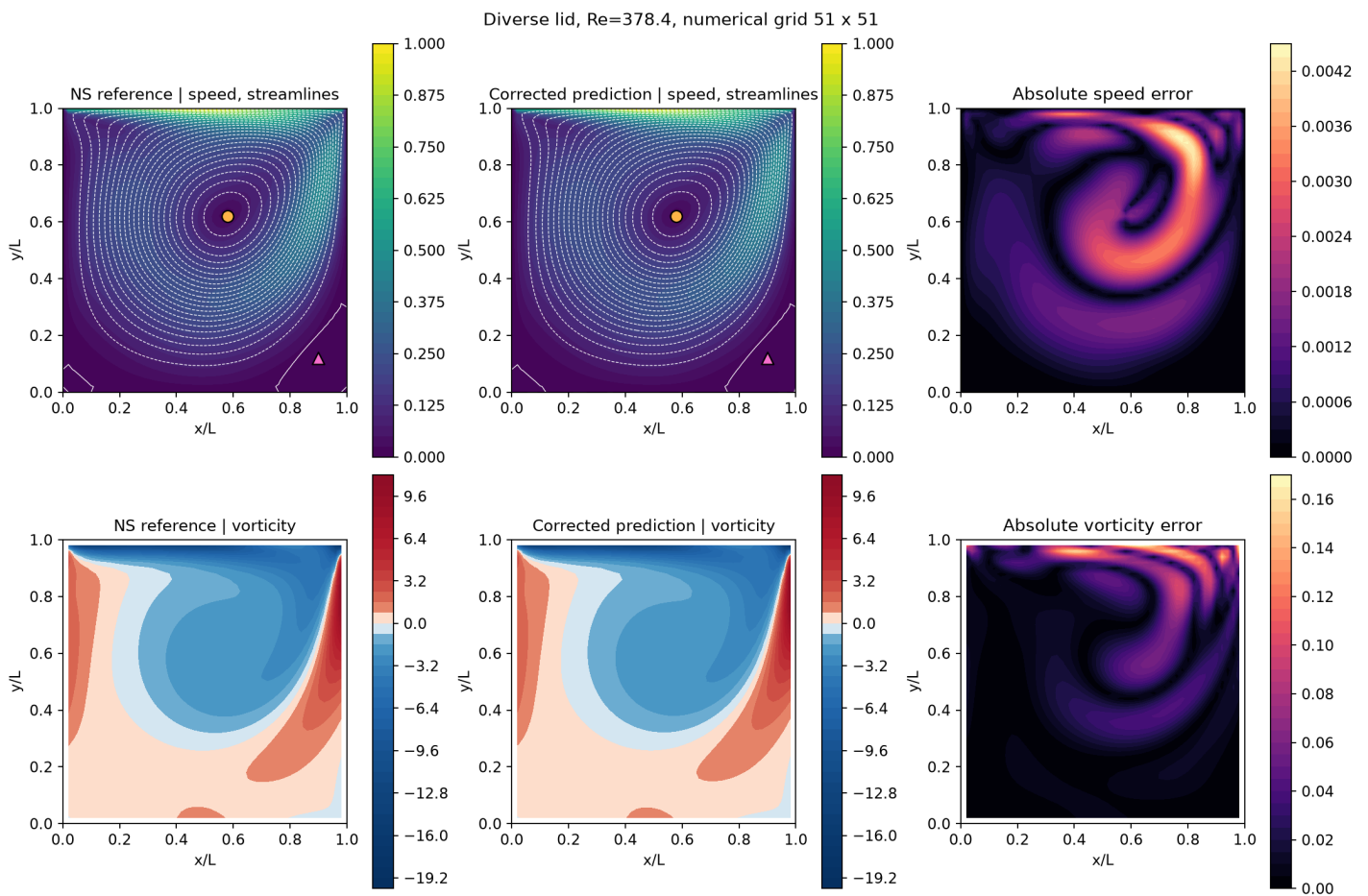
Week 4.2 | 2/3



Mean CFD 25-to-51 velocity change at coarse nodes: 20.59%

Vortex positions are measured on 51-node arrays; display contours do not add resolution.

One refinement step does not establish mesh-independent vortex capture.



Mean CFD 25-to-51 velocity change at coarse nodes: 17.81%

Vortex positions are measured on 51-node arrays; display contours do not add resolution.

One refinement step does not establish mesh-independent vortex capture.